

Multiplying matrices

Basic skills

Q1. The order $n \times m$ is n rows by m cols

Q2. A is order 2×2

B is order 3×2

C is order 3×3

TO conform
 $n \times m \times m \times l$
same

so BA, CB conform

Q3. a) $(1 \ 5) \begin{pmatrix} 3 \\ 2 \end{pmatrix} = 1 \times 3 + 5 \times 2 = 13$

b) $(2 \ -4) \begin{pmatrix} 3 \\ 2 \end{pmatrix} = 2 \times 3 + -4 \times 2 = -2$

c) $\begin{pmatrix} 1 & 5 \\ 2 & -4 \end{pmatrix} \begin{pmatrix} 3 \\ 2 \end{pmatrix} = \begin{pmatrix} 13 \\ -2 \end{pmatrix}$

Q4. $\begin{pmatrix} 1 & 5 \\ 2 & -4 \end{pmatrix} \begin{pmatrix} 3 & -2 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} 13 & -2 + 5 \\ -2 & -4 - 4 \end{pmatrix}$
 $= \begin{pmatrix} 13 & 3 \\ -2 & -8 \end{pmatrix}$
as in Q3.

Competency skills

Q5. $A = \begin{pmatrix} 5 & 3 \\ 6 & -1 \end{pmatrix}$ $B = \begin{pmatrix} 0 & -3 \\ 4 & 2 \end{pmatrix}$

$$AB = \begin{pmatrix} 0 + 12 & -15 + 6 \\ 0 - 4 & -18 - 2 \end{pmatrix} = \begin{pmatrix} 12 & -9 \\ -4 & -20 \end{pmatrix}$$

$$BA = \begin{pmatrix} 0 - 18 & 0 + 3 \\ 20 + 12 & 12 - 2 \end{pmatrix} = \begin{pmatrix} -18 & 3 \\ 32 & 10 \end{pmatrix}$$

SO $AB \neq BA$

Q6. a) AB as 2×2 and 3×2
Not conforming, not possible

b) BA as 3×2 and 2×2 conform

$$BA = \begin{pmatrix} 2 & -1 \\ 4 & 0 \\ 3 & -1 \end{pmatrix} \begin{pmatrix} 3 & 4 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 6 & 6 \\ 12 & 16 \\ 9 & 10 \end{pmatrix}$$

c) CB as 3×3 and 3×2 conforms

$$CB = \begin{pmatrix} 5 & 3 & 4 \\ 2 & 1 & 2 \\ 3 & 1 & 1 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ 4 & 0 \\ 3 & -1 \end{pmatrix} = \begin{pmatrix} 34 & -9 \\ 14 & -4 \\ 13 & -4 \end{pmatrix}$$

d) DE as 3×3 and 2×3 doesn't conform

e) DC as 3×3 and 3×3 conforms

$$DC = \begin{pmatrix} 5 & 2 & -1 \\ 0 & 2 & 0 \\ 3 & 5 & 1 \end{pmatrix} \begin{pmatrix} 5 & 3 & 4 \\ 2 & 1 & 2 \\ 3 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 26 & 16 & 23 \\ 4 & 2 & 4 \\ 28 & 15 & 23 \end{pmatrix}$$

f) EB as 2×3 and 3×2 conforms

$$EB = \begin{pmatrix} 2 & -1 & 0 \\ 4 & 2 & 4 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ 4 & 0 \\ 3 & -1 \end{pmatrix} = \begin{pmatrix} 0 & -2 \\ 28 & -8 \end{pmatrix}$$

g) BE as 3×2 and 2×3 conforms

$$BE = \begin{pmatrix} 2 & -1 \\ 4 & 0 \\ 3 & -1 \end{pmatrix} \begin{pmatrix} 2 & -1 & 0 \\ 4 & 2 & 4 \end{pmatrix} = \begin{pmatrix} 0 & -4 & -4 \\ 8 & -4 & 0 \\ 2 & -5 & -4 \end{pmatrix}$$

Q7. $M = \begin{pmatrix} 3 & 1 \\ 0 & 1 \end{pmatrix}$

$$M^2 = \begin{pmatrix} 3 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 3 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 9 & 4 \\ 0 & 1 \end{pmatrix}$$

$$M^3 = \begin{pmatrix} 3 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 9 & 4 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 27 & 13 \\ 0 & 1 \end{pmatrix}$$

$$\text{So } M^n = \begin{pmatrix} 3^n & (1+3+3^2+\dots+3^{n-1}) \\ 0 & 1 \end{pmatrix}$$

$$\text{Q8. } \begin{pmatrix} x & 0 & 4 \\ 2x & 1 & 3 \\ -2 & 5 & 3-x \end{pmatrix} \underbrace{\begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}}_{2I_3} \begin{pmatrix} x & -1 & 0 \\ 1 & x & 1 \\ 0 & -1 & x \end{pmatrix}$$

$$2 \begin{pmatrix} x & 0 & 4 \\ 2x & 1 & 3 \\ -2 & 5 & 3-x \end{pmatrix} \begin{pmatrix} x & -1 & 0 \\ 1 & x & 1 \\ 0 & -1 & x \end{pmatrix}$$

$$= 2 \begin{pmatrix} x^2 & -x-4 & 4x \\ 2x^2+1 & -x-3 & 3x+1 \\ 5-2x & 6x-1 & 5+3x-x^2 \end{pmatrix}$$

$$= \begin{pmatrix} 2x^2 & -2x-8 & 8x \\ 4x^2+2 & -2x-6 & 6x+2 \\ 10-4x & 12x-2 & 10+6x-2x^2 \end{pmatrix}$$

Advanced skills

Q9. a) $\begin{pmatrix} 3 & x \\ 1 & 4 \end{pmatrix}$ so $x=1$

same as symmetric

b) $M = \begin{pmatrix} -1 & x \\ x & 2 \end{pmatrix} \begin{pmatrix} 3 & 3 \\ -1 & 4 \end{pmatrix} = \begin{pmatrix} -3-x & -3+4x \\ 3x-2 & 3x+8 \end{pmatrix}$

For symmetry $3x-2 = -3+4x$

$$\Rightarrow -2 = -3+x \Rightarrow x=1$$

Q10.

$$M \begin{pmatrix} 4 & 2 & 1 \\ 10 & 2 & 3 \\ 8 & 5 & 4 \end{pmatrix} = \begin{pmatrix} 7 & 15 & 17 \end{pmatrix}$$

$1 \times 3 \quad 3 \times 3 \quad 1 \times 3$

must be 1×3 to conform

so $(a \ b \ c) \begin{pmatrix} 4 & 2 & 1 \\ 10 & 2 & 3 \\ 8 & 5 & 4 \end{pmatrix} = \begin{pmatrix} 7 & 15 & 17 \end{pmatrix}$

so

$$4a + 10b + 8c = 7 \quad a + 3b + 4c = 17$$

$$2a + 2b + 5c = 15$$

Solving simultaneous

$$a = 17 - 3b - 4c$$

$$4(17 - 3b - 4c) + 10b + 8c = 7$$

$$2(17 - 3b - 4c) + 2b + 5c = 15$$

$$\Rightarrow 68 - 2b - 8c = 7 \Rightarrow 2b + 8c = 61$$

$$34 - 4b - 3c = 15 \Rightarrow 4b + 3c = 19$$

Then $4b + 16c = 122$

$$\begin{array}{r} 4b + 16c = 122 \\ -4b + 3c = 19 \\ \hline \end{array}$$

$$13c = 103$$

$$\text{so } c = \frac{103}{13}$$

$$b = \frac{61 - 8 \times \frac{103}{13}}{2} = -\frac{31}{26}$$

$$a = 17 - 3 \times -\frac{31}{26} - 4 \times \frac{103}{13} = -\frac{289}{26}$$

$$\text{so } M = \frac{1}{26} \begin{pmatrix} -289 & -31 & 206 \end{pmatrix}$$

If it was SM then could use $M = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$
for $SM = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$